

Deep Learning

Beyond Deep Learning

Modern AI Systems and Intelligent Agents

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Adversarial Machine Learning



Adversarial Machine Learning

- **Adversarial Machine Learning (AML)** in the *simplest possible way*:
 - It's about **tricking AI systems on purpose**.
 - Imagine you trained an AI to recognize cats 🐱 and dogs 🐶.
 - Normal image → AI says: "This is a cat ✅"
 - But someone makes a **tiny change** (like invisible noise pixels)
 - Now the AI might say: "This is a dog ❌"
 - Even though *you* still see a cat.
 - **What's happening?**
 - Someone is **intentionally trying to fool the AI**.
 - That's called **adversarial (attacker vs AI) behavior**.
 - **Adversarial Machine Learning = studying how AI can be attacked or fooled, and how to make it stronger.**
 - **Real-life intuition**
 - Think of it like:
 - AI = student taking a test
 - Adversarial attack = trick questions designed to confuse the student
 -
- ## Why it matters
- Because AI is used in:
 - Self-driving cars 🚗
 - Medical diagnosis 🏥
 - Face recognition 📷
- If someone can fool AI → it can be **dangerous**.
- ## Simple summary
- AI learns patterns
 - Attackers exploit weaknesses
 - Slight changes → wrong prediction
 - Goal: **understand attacks & defend against them**



Adversarial Machine Learning

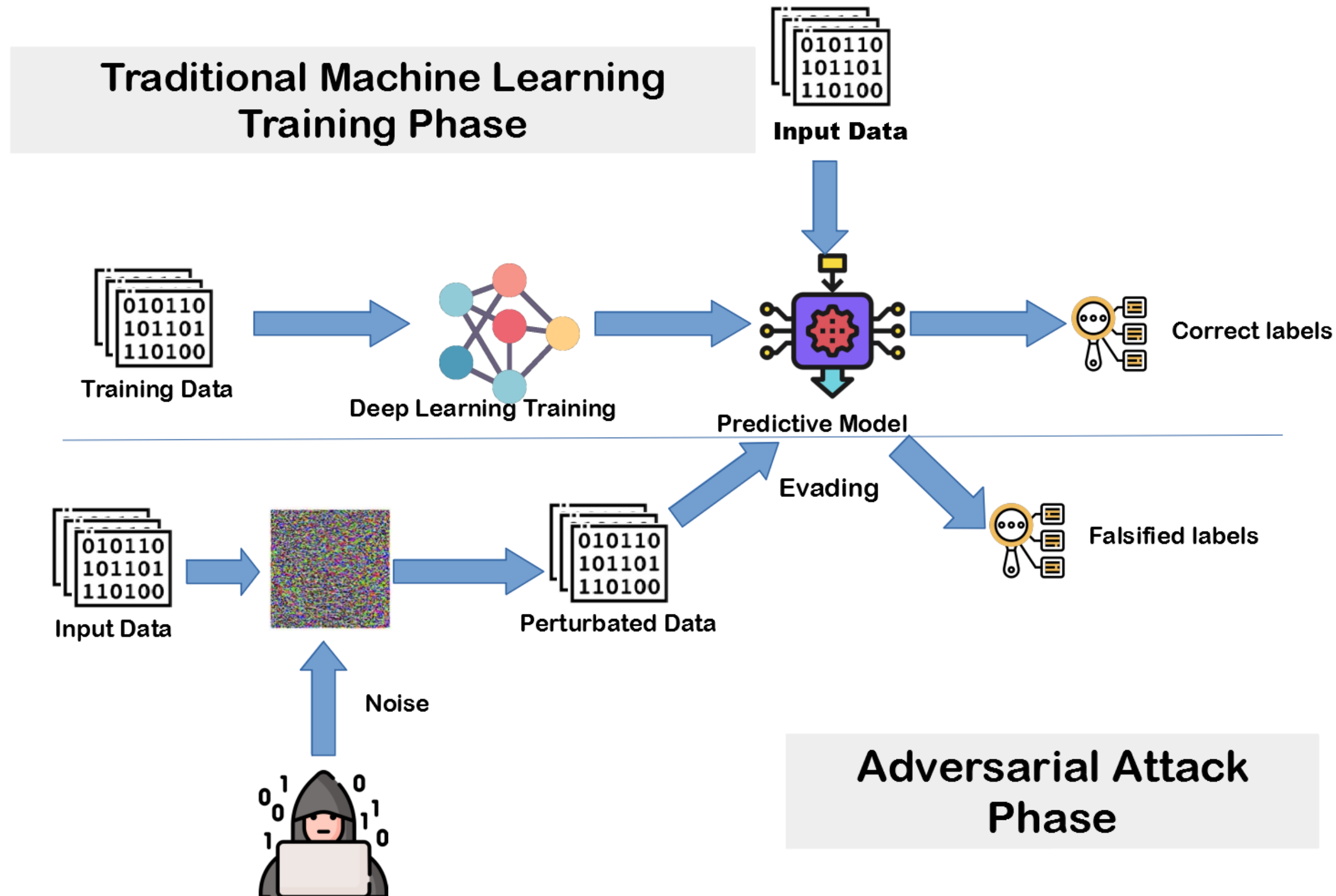
- Deep learning models can achieve very high accuracy on benchmark datasets, but this does not automatically mean they are reliable in hostile / manipulated settings.
- In many real-world applications, inputs are not always benign. An attacker may intentionally manipulate an image, audio signal, or text input.
- A model that works well on clean test data may fail when the input is carefully slightly modified.
- This matters in domains such as:
 - autonomous driving
 - face recognition
 - medical AI
 - spam filtering
 - malware detection
 - financial fraud detection
- Adversarial Machine Learning studies:
 - how models are attacked
 - why they are vulnerable
 - how to make them more robust
- Modern deep learning is powerful, but not automatically trustworthy.
- Accuracy on a standard test set is only one part of the story. If an intelligent adversary is present, the model may behave very differently.
- That is the motivation for adversarial machine learning.



What Is an Adversarial Example?

- An adversarial example is an input intentionally modified to make a machine learning model produce an incorrect prediction.
- Usually, the attacker starts with a normal input x and creates a modified input x_{adv} .
- The modification is designed to be:
 - small
 - hard for humans to notice
 - highly effective against the model
- Formally, we often think of:
 - $x_{adv} = x + \delta$
 - where δ is a small perturbation
- Example:
 - a panda image still looks like a panda to us
 - model predicts a completely different class

What Is an Adversarial Example?

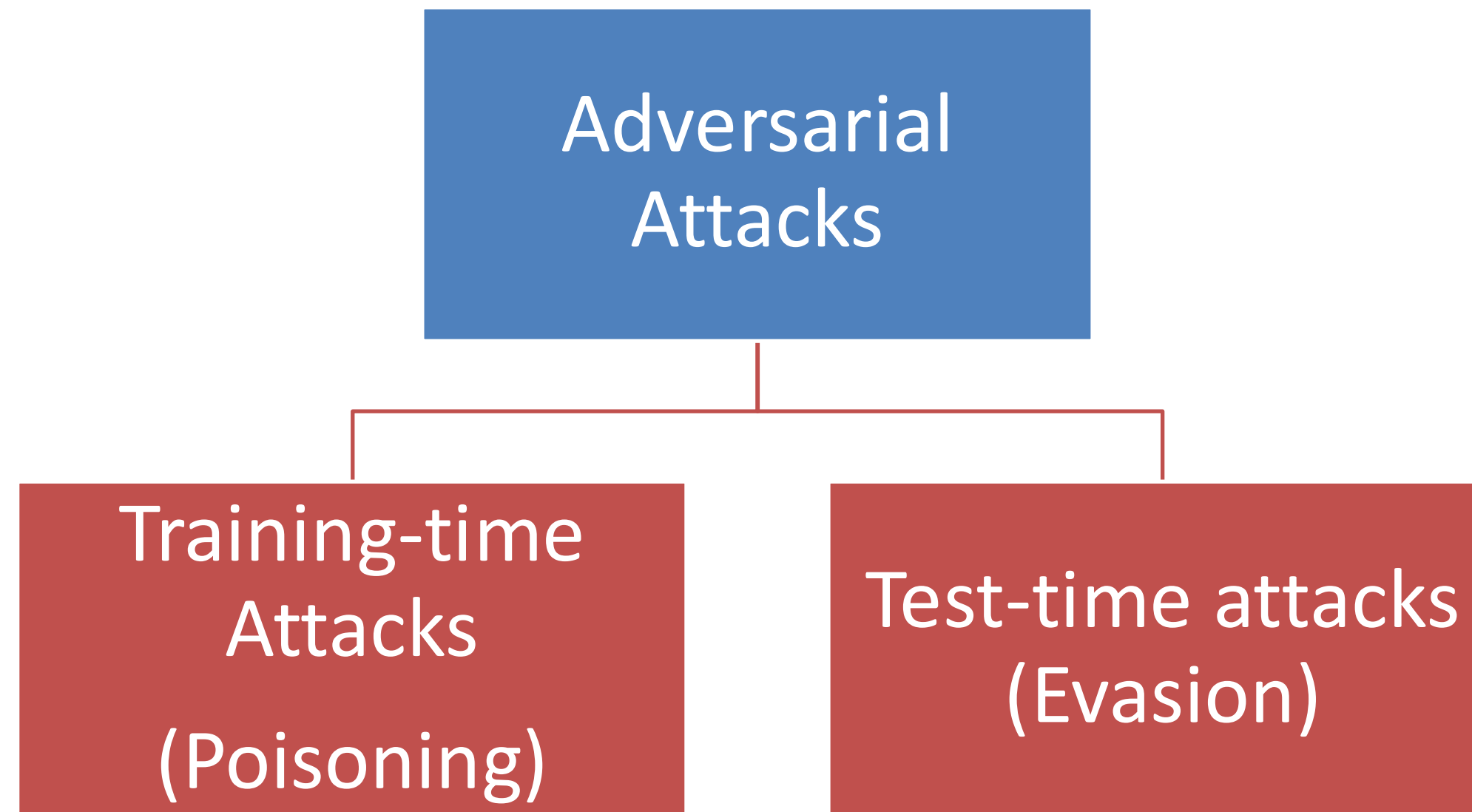


Why Can Tiny Changes Fool Deep Networks?

- Several intuitions explain this surprising behaviour:
- Deep networks operate in very high-dimensional spaces
- Even a tiny change in each dimension can combine into a large overall effect
- Decision boundaries learned by deep models can be close to data points in unexpected directions
- Models may rely on patterns that are predictive for training data but not robust to manipulation
- Humans and models do not “see” the same features:
 - humans focus on semantic meaning
 - models may depend on fragile statistical cues
- Therefore:
 - visually tiny changes can shift the sample across a decision boundary
 - confidence can remain high even when the prediction becomes wrong



The Core Security Question



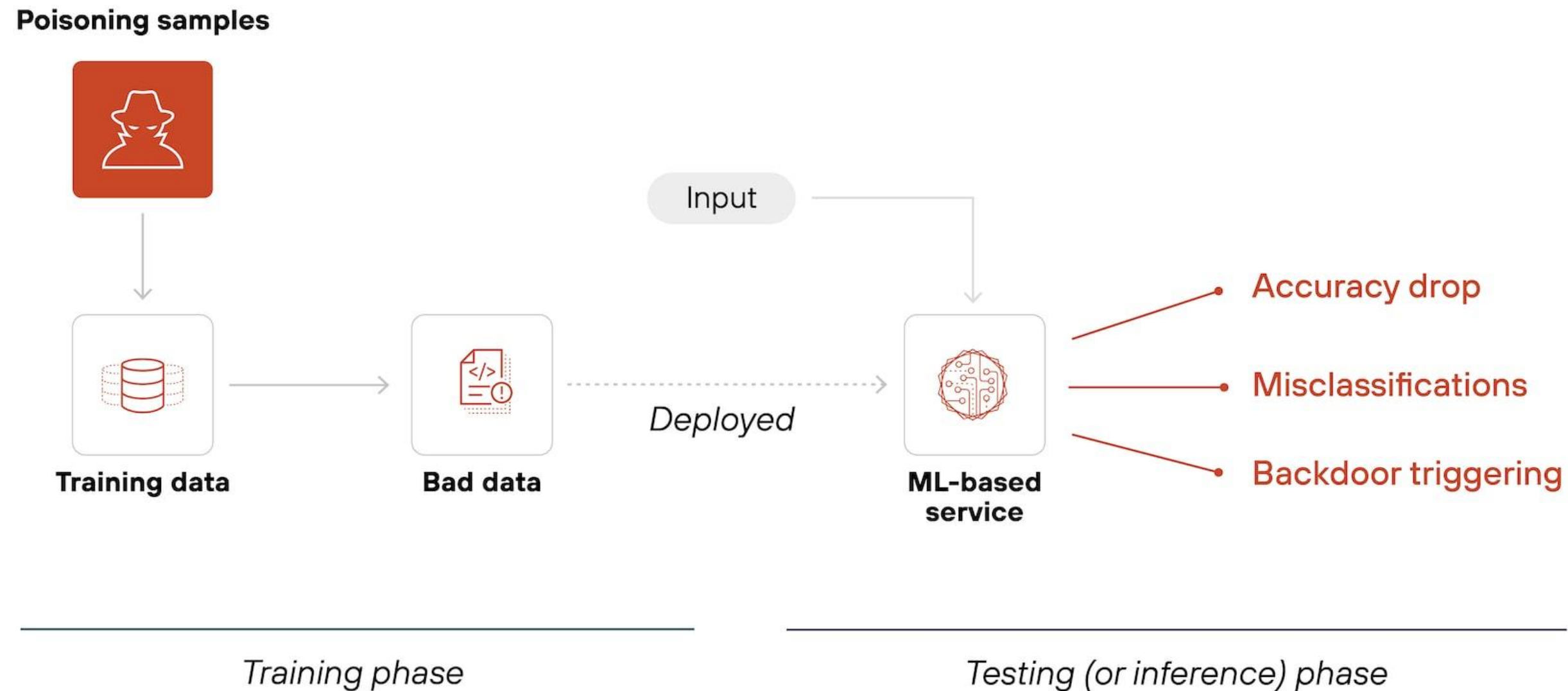
Training-Time Attacks (Poisoning Attacks)

- The attacker **interferes** with training data
- Goal: **Make the model learn wrong things**
- Think of a student:
 - If you give WRONG notes
 - Student learns WRONG concepts
 - Same with AI
- **How attacker does it**
- Adds fake data
- Changes labels (cat → labeled as dog)
- Injects misleading samples
- **Example**
- You train a spam classifier:
- **Normal:** “Win money now!!!” → spam
- Attacked training data:
 - “Win money now!!!” → NOT spam ❌
- Model learns wrong rule → fails later
- **Objective**
- **Corrupt learning process**
- Attack happens **before model is trained**
- Model itself becomes **weak from inside**



Training-Time Attacks (Poisoning Attacks)

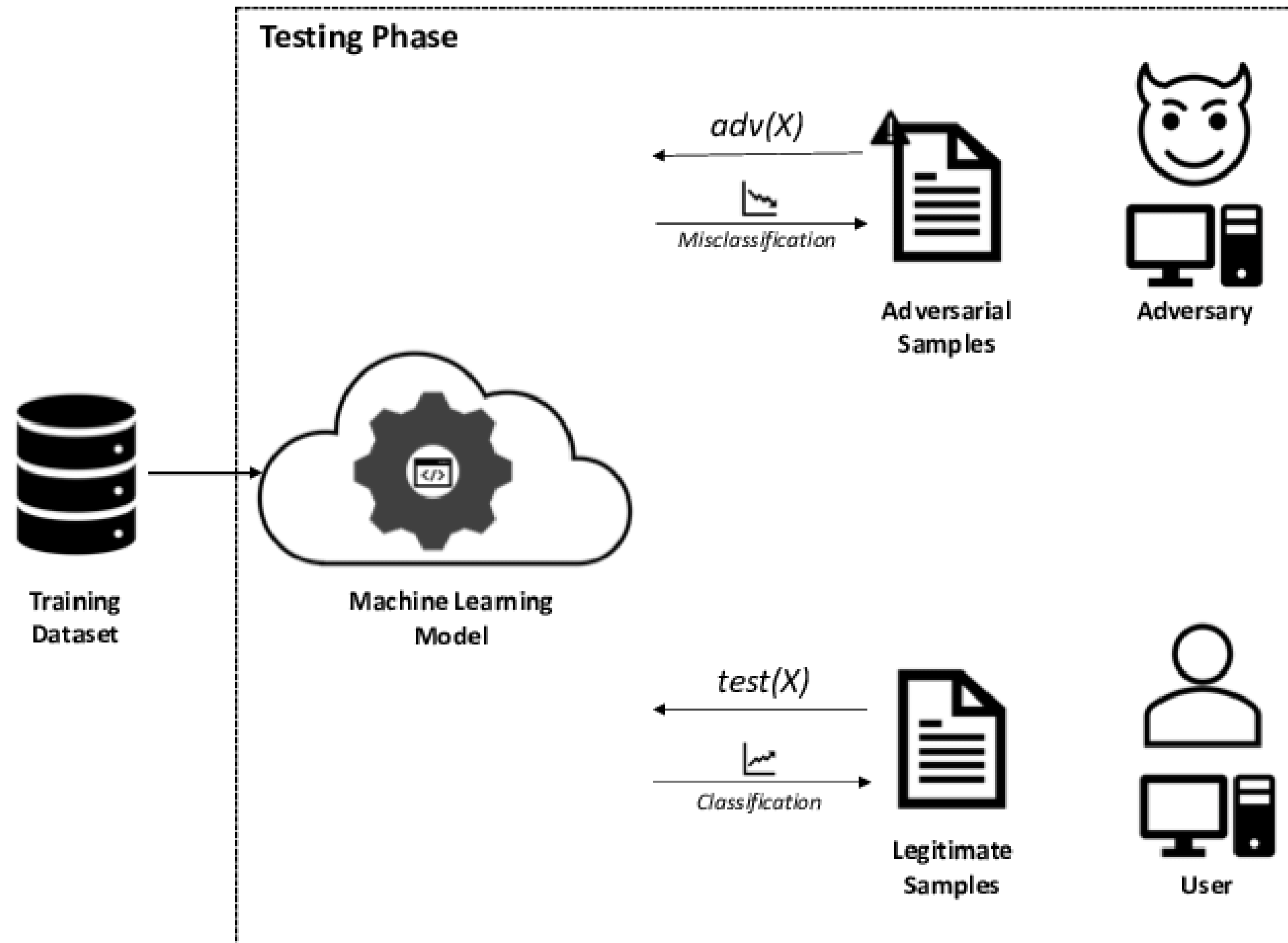
Data poisoning attack



Test-Time Attacks (Evasion Attacks)

- The model is already trained
- Attacker **does NOT change model**
- But **changes input at prediction time**
- **Intuition**
- Think of exam:
 - Student is well-trained
 - But question is tricky or misleading
 - Student gives wrong answer
- **How attacker does it**
 - Slightly modify input
 - Add small noise
 - Change words / pixels
- **Example (image)**
- Original: Stop sign → AI says “STOP” 
- Add tiny noise → AI says “Speed limit” 
- **Example (NLP - your area)**
 - Original: “This movie is great” → Positive 
 - Attack: “This movie is gr8” or slight word tricks
 - → AI may misclassify
- **Objective**
- **Fool model at inference time**
- Model **unchanged**
- Input is **manipulated**

Test-Time Attacks (Evasion Attacks)



Defenses at Training-Time (against poisoning)

- Problem: attacker corrupts training data
- **1. Data Cleaning (Sanitization)**
 - Check data before training
 - Remove suspicious / noisy samples
- **Example:**
 - Detect outliers
 - Remove wrong labels
- Idea:
 - “Don’t trust all data blindly”
- **2. Robust Training Algorithms**
- Use models that are less sensitive to noise
- Example:
 - Regularization
 - Robust loss functions
- Idea:
 - “Even if bad data exists, model stays stable”
- **3. Data Validation & Trusted Sources**
 - Use verified datasets
 - Avoid random internet data
- Idea:
 - “Use clean, reliable data”
- **4. Adversarial Training (also works here)**
 - Add some **poisoned examples** intentionally
 - Train model to handle them
- Idea:
 - “Train with attacks → become stronger”



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Defenses at Test-Time (against evasion)

- Problem: attacker manipulates input
- **1. Adversarial Training (MOST IMPORTANT)**
- Train model on:
 - Normal data
 - Attacked data
- Result: Model learns: “Don’t get fooled by small tricks”
- **2. Input Preprocessing**
- Clean the input before giving to model
- Example:
 - Remove noise
 - Normalize text
 - Spell correction (for NLP)
- Idea:
 - “Fix input before prediction”
- **3. Detection Mechanisms**
- Detect if input looks suspicious
- Example:
 - If input is too weird → reject it
- Idea:
- “Catch attacker at the door”
- **4. Model Ensemble**
- Use multiple models
- If one is fooled → others can correct
- Idea:
 - “Multiple opinions reduce risk”
- **5. Gradient Masking / Defensive Techniques**
- Make attacks harder
- Hide model behavior
- Idea:
 - “Make it difficult to attack”



If you remember only ONE thing

- **Adversarial Training = Best general defense**
- Why?
- It prepares model for real attacks
- Works for both training & testing scenarios
- **Simple analogy**
- Poisoning defense → *check ingredients before cooking* 🍲
- Evasion defense → *check food before eating* 🍴



Generative Adversarial Networks (GANs)



Generative Adversarial Networks (GANs)

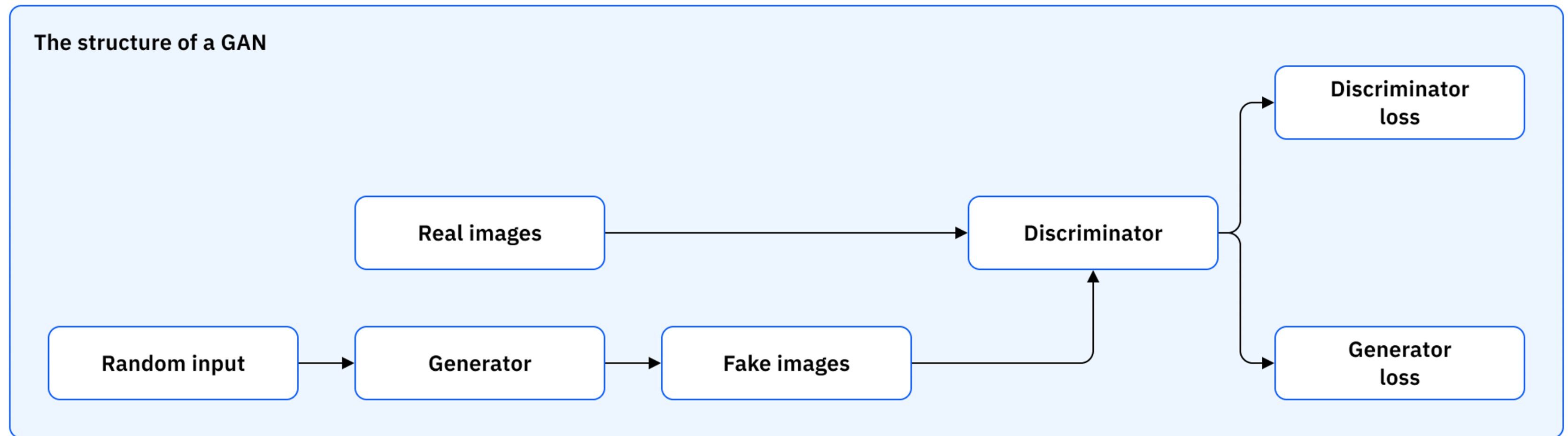
- Generative Adversarial Networks (GANs) are a machine learning framework where two neural networks
- **Generator - Fake**
- **Discriminator - Detect**
- Against each other to create new, realistic synthetic data, such as images, video, or audio



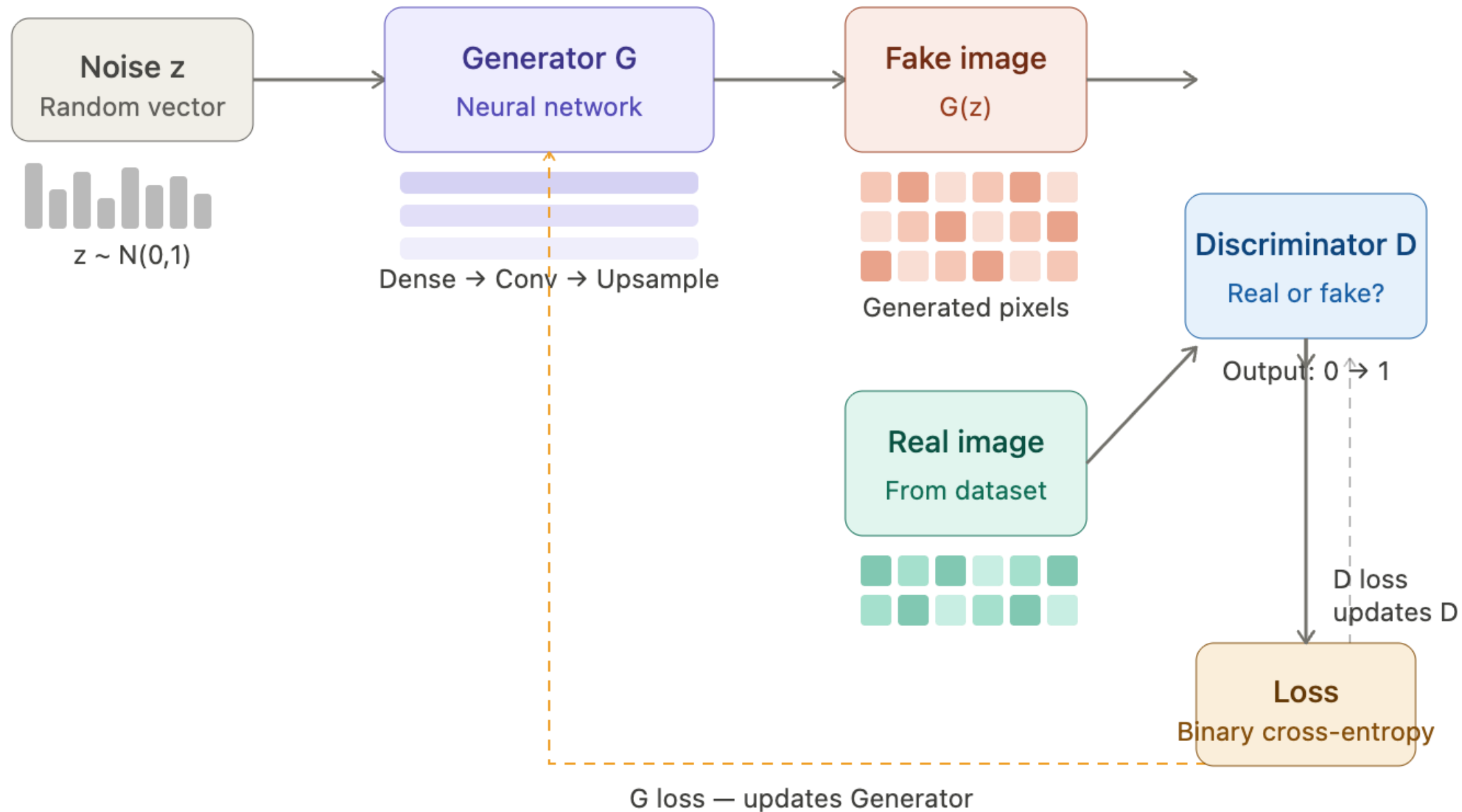
Two players in (GANs)

- **1. Generator (G)**
 - Creates fake data
 - Example:
 - Fake images of faces
 - Fake text
 - Fake signals
 - Goal:
 - “Make fake data look real”
- **2. Discriminator (D)**
 - Judges data
 - Input: real OR fake
 - Output: “real  ” or “fake  ”
 - Goal:
 - “Detect fake data correctly”

Two players in (GANs)



GAN pipeline for Images



Click any component to learn more

GAN pipeline for Images

- Detailed GAN pipeline for images, including:
- How noise works
- Models used
- Training algorithm
- Step-by-step from start to finish



1. Components

- **Generator (G)**
 - Input: Random noise vector z
 - Output: Fake image
 - Model used:
 - Deep Neural Network
 - (usually **DCGAN Convolutional Generator**)
- **Discriminator (D)**
 - Input: Image (real OR fake)
 - Output: Probability (real = 1, fake = 0)
 - Model used:
 - CNN (Convolutional Neural Network)

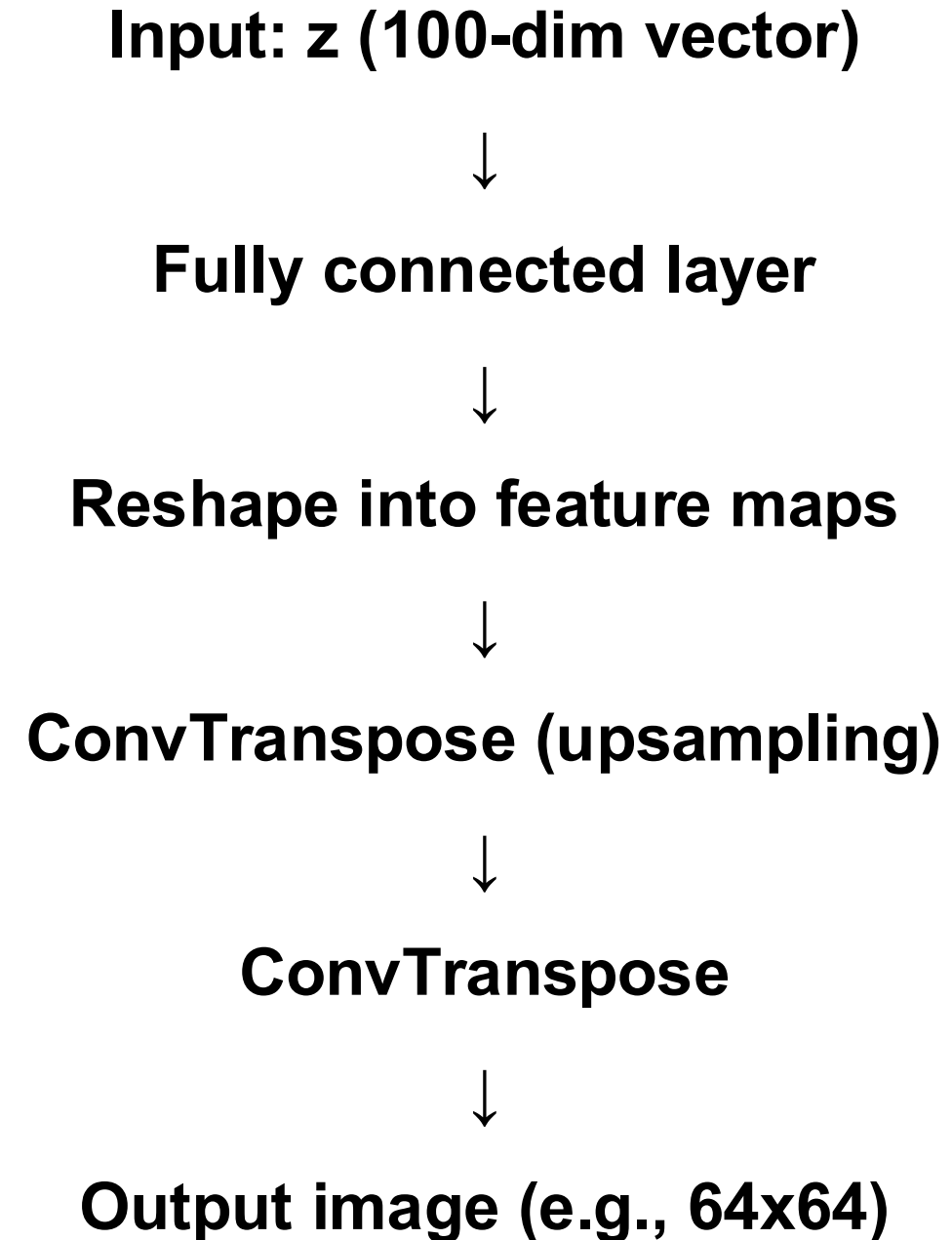
What is “Noise” here?

- **Noise definition**
 - $z \sim \text{Normal}(0,1)$
- Example:
- $z = [0.2, -1.1, 0.5, \dots, 0.8]$ (size = 100)
- This vector has **no meaning initially**
- **Intuition**
- Noise = **random seed**
- Generator learns:
- “How to convert random noise $\rightarrow$ meaningful image”.



3. Model Architectures (DCGAN)

- Generator (G)



- Output example: Fake face image

3. Model Architectures (DCGAN)

- Discriminator (D)

Input: image (64x64)



Conv layer



Conv layer



Flatten



Fully connected



Sigmoid

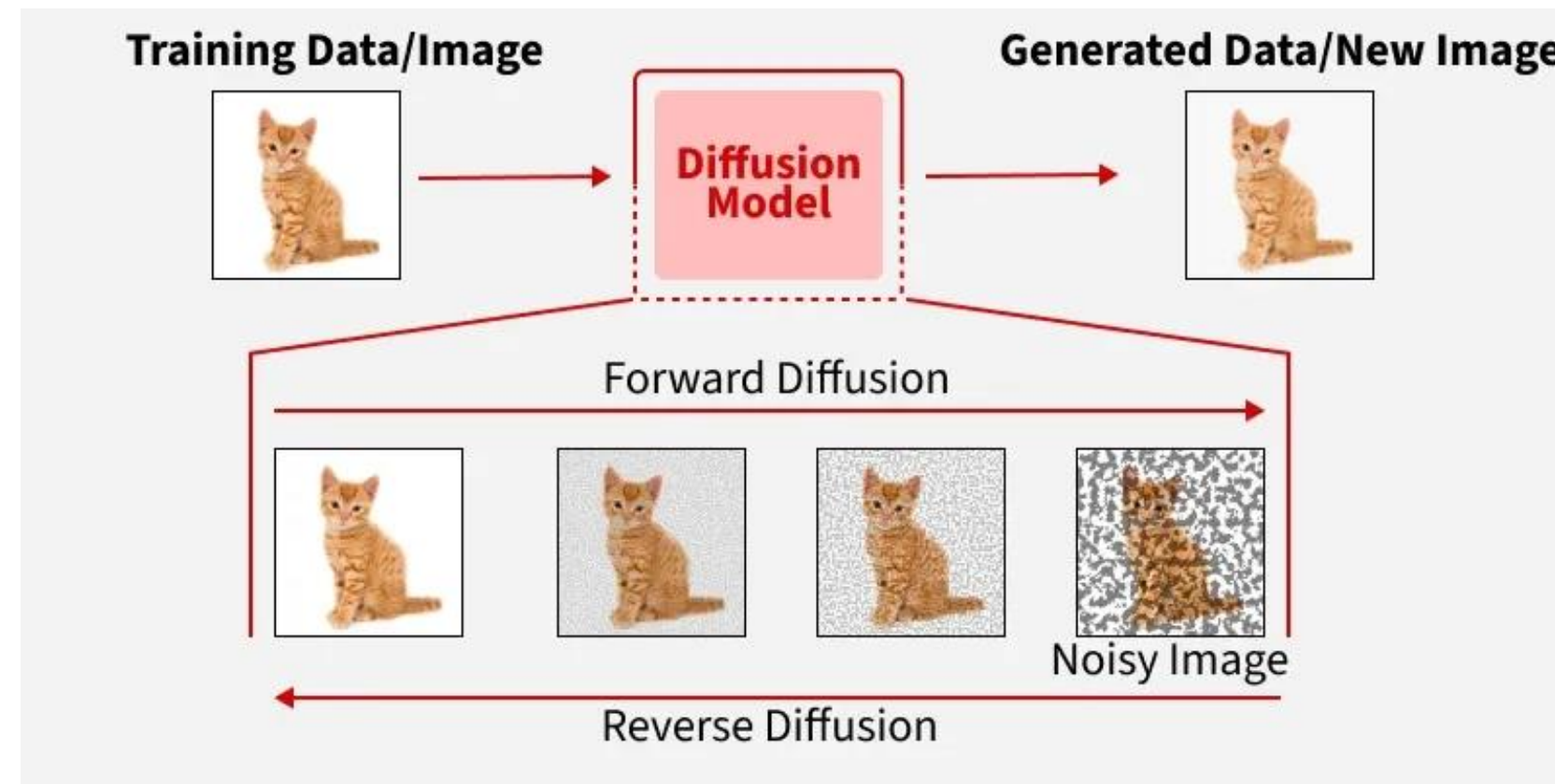
Output:

0.95 → real

0.10 → fake

Diffusion Models

- **Diffusion models** are powerful generative AI frameworks that create high-quality images, videos, and audio by learning to reverse a gradual noising process.
- They work by destroying training data with Gaussian noise (forward process) and training a neural network to reconstruct the data by removing this noise iteratively (reverse process).



Diffusion Models

- Diffusion Model = start from noise → slowly turn it into an image
- **Step 1: Forward Process (NOISING)**
- Take a real image
- Gradually add noise
- **Example:**
 - Step 0: Original image (digit 5)
 - Step 1: Slight noise
 - Step 2: More noise
 - Step 3: Very noisy
 - ...
 - Step T: Pure random noise
- End result:
 - Image → Random noise
 - **Important**
 - This process is: Fixed
 - No learning
- **Step 2: Reverse Process (LEARNING PART)**
- Now the model learns:
 - “How to remove noise step-by-step”
- **Process:**
 - Noise → less noise → clearer → image



What does the model learn?

- Instead of generating image directly, it learns:
- Predict the noise in the image
- **Example:**
- Input:
 - Noisy image
- Model predicts:
 - Noise part
- Then:
 - Remove noise → cleaner image
- **. Full Generation Process**
- After training:
- **Step 1: Start with noise**
- x_T = pure random noise
- **Step 2: Run reverse process**
- $x_T \rightarrow x_{T-1} \rightarrow x_{T-2} \rightarrow \dots \rightarrow x_0$
- **Step 3: Final result**
- x_0 = clean image 
- **GAN:**
 - “Draw image in one shot”
- **Diffusion:**
 - “Start messy → slowly clean it”

Reinforcement Learning



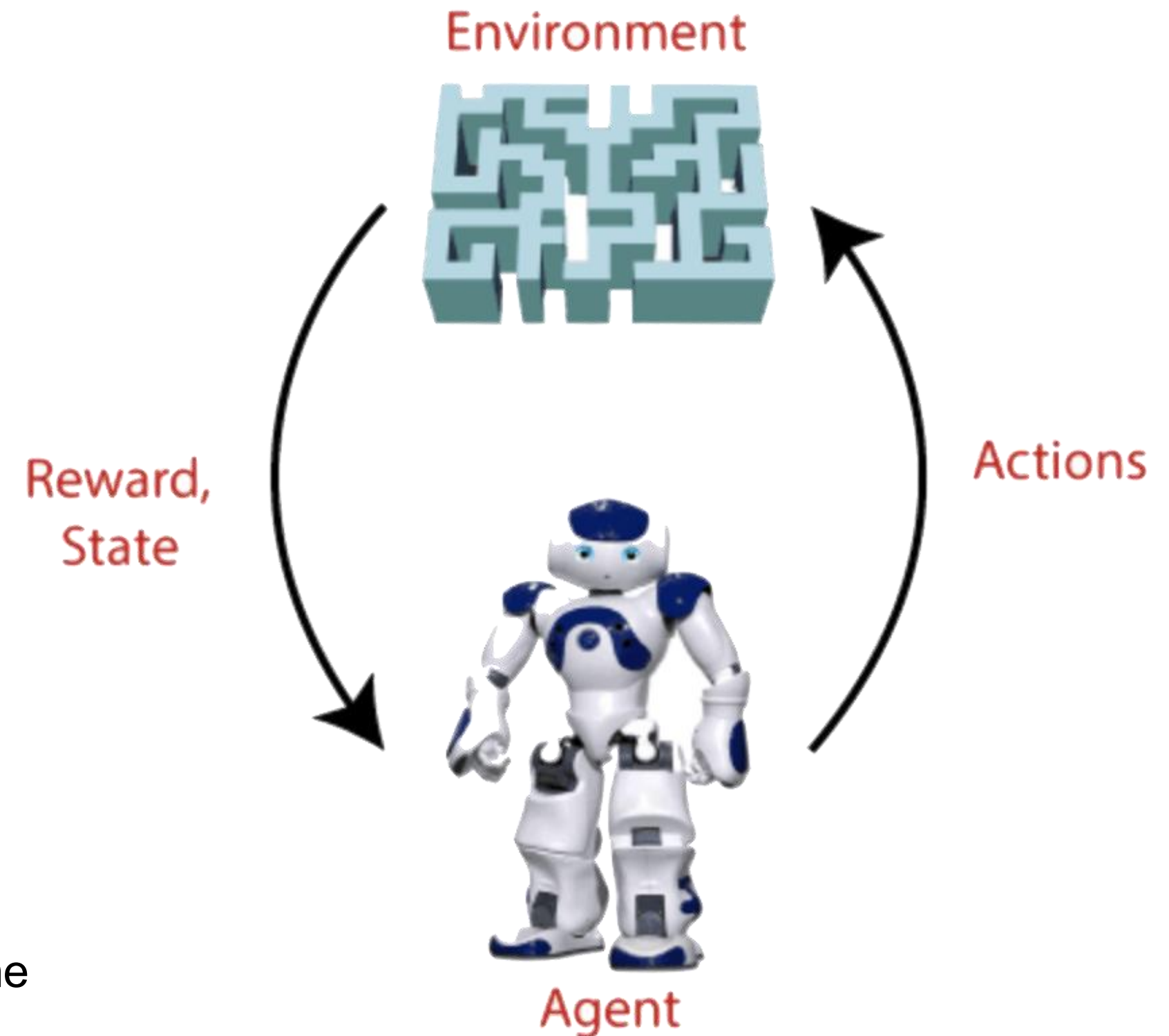
What is Reinforcement Learning?

- **Reinforcement Learning** = learning by trying, getting feedback, and improving decisions over time.
- **Agent** = any decision-making program/system
- **Agent** = the entity that makes decisions
- An agent in Reinforcement Learning is not a robot; it is any system (software or hardware) that takes actions and learns from feedback.
- It can be:
 - A robot
 - A software program
 - A game player
 - A recommendation system
 - Even LLM ChatGPT-like systems



What is Reinforcement Learning?

- Reinforcement learning is a type of machine learning method where an intelligent agent interacts with the environment and learns to act within that.
- For each good action, the agent gets positive feedback, and for each bad action, the agent gets negative feedback or penalty.
- RL solves a specific type of problem where decision making is sequential, and the goal is long-term.
- The agent continues doing following three things:
 - Take action,
 - change state/remain in the same state
 - get feedback
- By doing these actions, he learns and explores the environment.



Examples

- **1. Game**

- Agent = game AI
- Environment = game screen

- Example:

- Agent decides: move left, jump, attack

- **2. YouTube Recommendation**

- Agent = recommendation algorithm
- Environment = user behavior

- Action:

- Show video A or B

- Reward:

- User clicks → good
- User ignores → bad

- **3. Self-driving car**

- Agent = driving system
- Environment = road

- Here hardware exists (car), but RL is still about decisions

- **4. Robot Agent = robot**

- Environment = physical world



Examples

- **Step 1: Initial State**

- The process always starts with a **state**
- S_0 = current situation
- **Example:**
- Game → player at position (x, y)
- YouTube → user watching video
- Car → speed = 50 km/h

- **Step 2: Agent Observes State**

- The agent “looks” at the environment
- Agent sees state S_0
- It understands: “Where am I right now?”

- **Step 3: Agent Chooses Action**

- Based on state, agent takes an action
- A_0 = action chosen in S_0
- **Example:**
- Move left
- Show video A
- Apply brake

- **Step 4: Action is Applied to Environment**

- Environment changes due to action
- $S_0 + A_0 \rightarrow$ Environment transition

- **Step 5: New State Generated**

- After action, environment moves to next state
- S_1 = new state
- **Example:**
- Character moves position
- User clicks video
- Car slows down

- **Step 6: Reward is Given**

- Environment gives feedback
- R_1 = reward
- **Example:**
- +10 → good action
- -5 → bad action
- 0 → neutral

- **Step 7: Agent Learns from Experience**

- Agent updates its knowledge
- (S_0, A_0, R_1, S_1)
- Learns: “Was this action good or bad?”

- **Step 8: Move to Next Step (Repeat)**

- Now: Current state = S_1
- Repeat again: $S_1 \rightarrow A_1 \rightarrow S_2 \rightarrow R_2 \rightarrow$ learning



Supervised vs. Unsupervised vs. Reinforcement Learning

Criteria	Supervised ML	Unsupervised ML	Reinforcement ML
Definition	Learns by using labelled data	Trained using unlabelled data without any guidance.	Works on interacting with the environment
Type of problems	Regression and classification	Association and Clustering	Exploitation or Exploration
Algorithms	Linear Regression, Logistic Regression, SVM, KNN etc.	K – Means, C – Means, Apriori	Q – Learning, SARSA
Aim	Calculate outcomes	Discover underlying patterns	Learn a series of action
Application	Risk Evaluation, Forecast Sales	Recommendation System, Anomaly Detection	Self Driving Cars, Gaming, Healthcare



Term Used in Reinforcement Learning

Agent

→ An entity that can perceive/explore the environment and act upon it.

Environment

→ A situation in which an agent is present or surrounded by.

Action

→ Actions are the moves taken by an agent within the environment.

State

→ State is a situation returned by the environment after each action taken by the agent.

Reward

→ A feedback returned to the agent from the environment to evaluate the action of the agent.

Policy

→ Policy is a strategy applied by the agent for the next action based on the current state.

Value

→ It is expected long-term return with the discount factor and opposite to the short-term reward.

Agentic AI



What is Agentic AI?

- **Agentic AI** = AI systems that can **take actions**, **make decisions**, and **complete tasks** autonomously using **tools and reasoning**.
- **Why “Agentic”?**
- Because it behaves like an **agent**:
- Observe → Think → Act → Repeat
- Instead of just answering: It **does tasks**



Core Components

- **1. LLM (Brain)**
- The reasoning engine
- GPT-like model
- It can:
 - Understand task
 - Plan steps
 - Decide actions
- **2. Tools (Hands)**
- External functions the agent can use:
- Examples:
 - Search web
 - Send email
 - Execute code
 - Access database
- **3. Memory (storage)**
- Stores past information
- Short-term (conversation)
- Long-term (knowledge)
- **4. Environment**
- Where it operates:
 - User input
 - Internet
 - APIs
 - Files
- **5. Action Module**
- Executes decisions:
- Call API
- Run code
- Generate output



Example

- **Task:**
 - "Find top AI papers and summarize them"
- **Step 1: Observe**
 - User request
- **Step 2: Think**
 - **Need to:**
 - 1. Search papers
 - 2. Read content
 - 3. Summarize
- **Step 3: Act**
 - Call search API
- **Step 4: Observe result**
 - List of papers
- **Step 5: Act again**
 - Read papers
- **Step 6: Final output**
 - Summary
- This is **agent behavior**, not simple response



Architecture (Modern Systems)

- **Example Frameworks:**

- LangChain

- AutoGPT

- OpenAI Agents

- **Typical architecture:**

- **User → LLM → Planner → Tool → Result → LLM → Final Output**



Planning (Important Feature)

- Agent doesn't just respond: it **plans steps**
- **Example:** Goal: Book a flight
- Plan:
 - 1. Search flights
 - 2. Compare prices
 - 3. Select best
 - 4. Book
- This is beyond normal ML
- **Iterative Reasoning (Loop)**
- Agent can retry:
 - try another method
- Makes it: Autonomous
- **Connection with RL**
- Earlier you learned RL:
- Agent ↔ Environment ↔ Reward
- Agentic AI is similar conceptually:

RL	Agentic AI
State	Context
Action	Tool use
Reward	success/failure
Policy	reasoning



Types of Agents

- **1. Reactive Agent**
 - Just responds
 - No planning
- **2. Planning Agent**
 - Creates multi-step plan
- **3. Tool-using Agent**
 - Uses APIs, tools
- **4. Autonomous Agent**
 - Works independently

Real-world Examples

- **ChatGPT (basic)**
 - mostly reactive
- **AutoGPT**
 - fully autonomous
 - multi-step execution
- **Browsing agents**
 - search + extract + summarize
- **Business agents**
 - data analysis
 - report generation



Why Agentic AI is powerful

- Can perform tasks (not just answer)
- Can use real-world tools
- Can act step-by-step
- Can automate workflows
- **Limitations**
 - Can make wrong decisions
 - Tool misuse
 - Requires control (guardrails)
 - Not always reliable
- **OLD AI:**
 - Input → Output
- **Agentic AI:**
 - Goal → Plan → Act → Observe → Repeat → Result



What is ReAct?

- **Original Paper: "ReAct: Synergizing Reasoning and Acting in Language Models"**
- Year: 2022
- **Authors:** Researchers from Google Research + Princeton
- **ReAct = Reasoning + Acting**
 - This paper introduced the idea: Reasoning (thoughts) + Acting (tool use) = better AI agents
- **ReAct is a method where an AI system first reasons (thinks step-by-step) and then takes actions (like using tools), repeatedly.**
- **Core Idea**
- Instead of just answering directly:
 - Input → Output ❌
- ReAct does:
 - Think → Act → Observe → Think → Act → ...



Chain-of-Thought (CoT)

- Think step-by-step → answer

- Example:

- Step 1: ...
- Step 2: ...
- Answer: ...

- Only reasoning

No actions/tools

- **ReAct (extension of CoT)**

- Think → Act → Observe → Think → Act → ...

- Reasoning + tool usage

Feature	Chain-of-Thought	ReAct
Reasoning	✓	✓
Tool use	✗	✓
Interaction	✗	✓
Agent behavior	✗	✓



Why ReAct was needed?

- **Earlier LLMs:**
 - Could **reason** (like chain-of-thought)
 - Could **use tools** (like API calls)
- **BUT:**
 - Not together properly
- ReAct combines both:
 - Reasoning + Tool usage
- **2. ReAct Loop**
- This is the heart of it:
 - 1. Thought (Reason)
 - 2. Action (Tool call)
 - 3. Observation (Result)
- **3. Structure Format**
- A ReAct agent works like this:
 - **Thought:** I need to search for something
 - **Action:** Search["query"]
 - **Observation:** result from search
 - **Thought:** Now I understand, next step...
 - **Action:** ...
 - **Final Answer:** ...



Example

- **Question:**
 - Who is the president of France in 2023?
- **ReAct Process:**
 - Thought: I should search for current president
 - Action: Search["President of France 2023"]
 - Observation: Emmanuel Macron is president
 - Thought: Now I know the answer
 - Final Answer: Emmanuel Macron
- **See:**
 - It **thinks**
 - Uses **tool**
 - Uses result to decide next step

Another Example (More Realistic)

- **Task:**
 - "What is the population of Pakistan and add 10 million?"
- **ReAct Steps:**
 - Thought: I need to find population first
 - Action: Search["Pakistan population"]
 - Observation: 240 million
 - Thought: Now add 10 million
 - Action: Calculator[240 + 10]
 - Observation: 250
 - Final Answer: 250 million
- This is **true agent behavior**

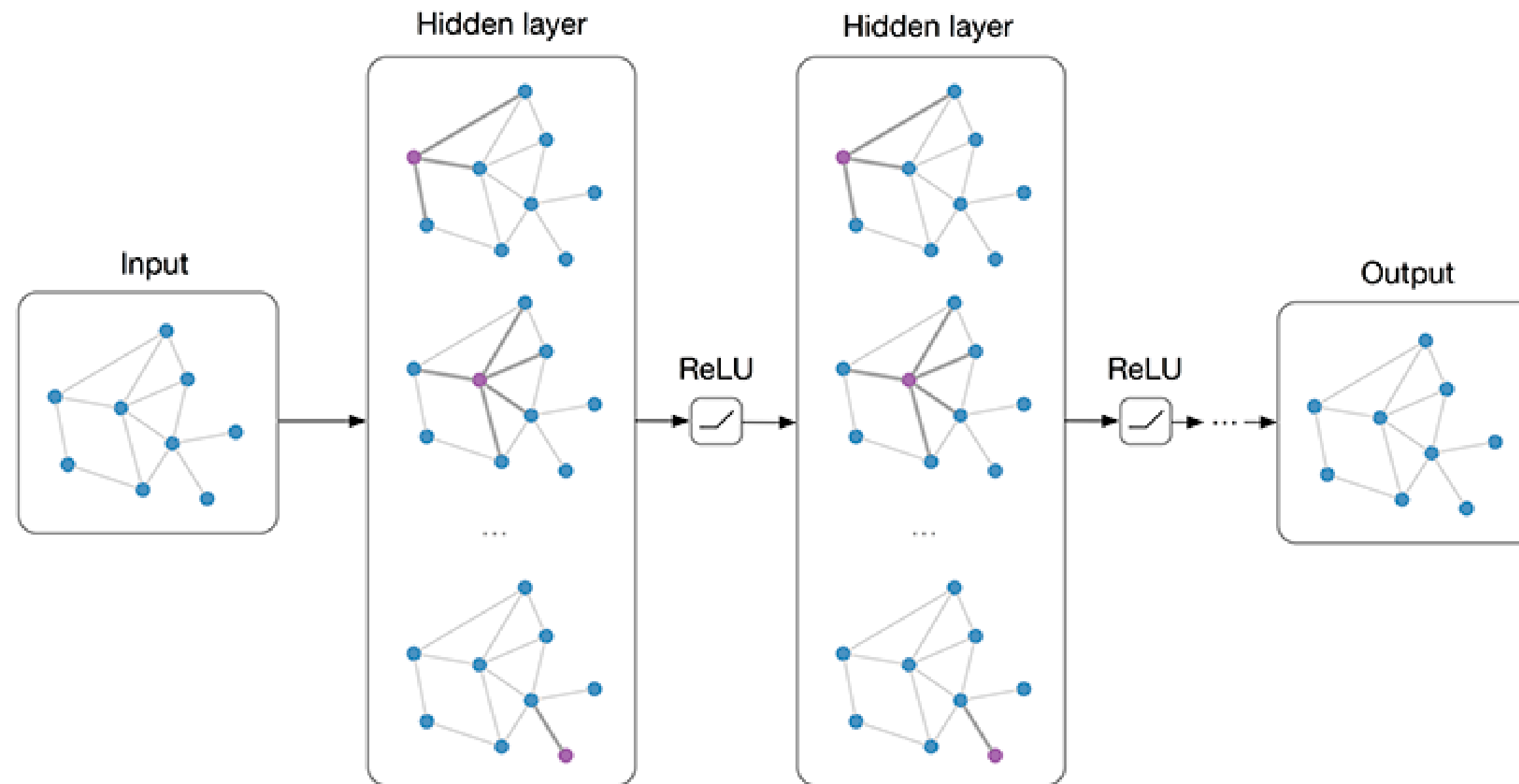


Graph Neural Network



Graph Neural Network (GNN)

- A Graph Neural Network (GNN) is a neural network designed to work on graph data (nodes + connections).



Graph Neural Network (GNN)

- Normal neural networks work on:
 - **Image:** pixels arranged in grid
 - **Text:** sequence of words
- But many real-world problems look like:
- Nodes connected to other nodes
- **Example:**
 - Social network 
 - Web pages 
 - Knowledge graphs 
 - Molecules 
- That structure = **Graph**

Graph Neural Network (GNN)

- **What is a Graph?**
- A graph has: $G = (\text{Nodes}, \text{Edges})$
- **Nodes (V)**
 - **Objects**
 - Example: people, words, atoms
- **Edges (E)**
 - **Connections**
 - Example: friendship, links, relationships

- **Example**

A — B — C

|

D

- A, B, C, D = nodes
- lines = edges



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Why Normal Neural Networks Fail?

- They expect **fixed structure**
- Graphs are:
 - irregular
 - variable size
 - connected
- So we need:
- **Core Idea of GNN**
- Each node learns from its neighbors
- Node representation = information from:
 - itself + neighbors
- **Message Passing**
 - This is the heart of GNN



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Step by Step:

- **Step 1: Initialize features**
- Each node has features:
 - $A = [1, 0]$
 - $B = [0, 1]$
 - $C = [1, 1]$
 - $D = [0, 0]$
- **Step 2: Collect neighbor information**
- Node B collects info from A, C, D
- **Step 3: Aggregate (combine neighbors)**
- $New_B = \text{average}(A, C, D)$

- **Step 4: Update node**
- $B_{new} = f(old_B + \text{neighbors})$
- This repeats many times

What does the model learn?

- After multiple layers:
 - Node understands:
 - local structure
 - global structure

Types of Tasks in GNN

- **1. Node Classification**

- Predict label of node

- Example:

- User → Fake or Real account

- **Edge Prediction**

- Predict connection

- Example:

- Will A become friend with B?

- **3. Graph Classification**

- Classify entire graph

- Example:

- Molecule is toxic or not

- **GNN Architecture**

- Input: Graph (nodes + edges)
- Message Passing Layers
- Node embeddings
- Prediction layer

Common GNN Models

- **1. GCN (Graph Convolutional Network)**

- Most basic
- Uses averaging

- **2. GAT (Graph Attention Network)**

- Uses attention
- decides:
 - which neighbor is more important

- **3. GraphSAGE**

- Samples neighbors
 - good for large graphs



Example (Social Network)

- **Nodes:**
 - Users
 - **Edges:**
 - Friendship
 - **Task:**
 - Detect fake accounts
 - GNN learns:
 - User behavior + neighbor behavior
 - Better prediction
- **Why GNN is powerful**
 - Uses relationships
 - Captures structure
 - Works on complex networks
 - Handles non-Euclidean data



Real-world Applications

- Social networks (friend suggestions)
- Recommendation systems
- Fraud detection
- Knowledge graphs (Google search)
- Molecule/drug discovery
- NLP (dependency trees)



Final Intuition

- Think:
 - CNN → looks at nearby pixels
 - GNN → looks at **connected nodes**
- Graph Neural Networks learn representations of nodes by aggregating information from their neighbors in a graph structure.
- **One-line takeaway**
- **GNN = learning from connections, not just data points**



Message to Students

Closing Message

Dr. Zohair Ahmed

Department of AI & Data Science

Deep Learning: A Journey Completed, A Future Begins



Message to Students

- **Before We End This Course...**
- Today is not just the end of a course
- It is a transition point in your journey
- *You came here to learn algorithms...*
- *But you are leaving with the ability to build intelligence*



Message to Students

- **What You Don't Know (Yet)**
 - You have not mastered AI
 - None of us have
- **Because:**
 - AI is evolving every month
 - New architectures appear constantly
 - Systems are becoming more complex
- *This field rewards learners, not memorizers*



Message to Students

- **From Student to Researcher**
- Until now:
 - You solved assignments
 - You followed instructions
 - You implemented known ideas
- From today onward:
 - You question everything
 - You explore unknowns
 - You create new ideas
- This shift defines your future



Message to Students

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Message to Students

- **A Practical Framework**
- Start with:
 - Take a known model
 - Find its limitation
 - Modify one component
 - Test experimentally
- Repeat.
 - Research = Iterative curiosity



Message to Students

- **Systems Over Models**
 - Future AI is not about training a model.
- It is about building:
 - Data pipelines
 - Model architectures
 - Memory components
 - Tool integration
 - Decision loops
- In short: **Complete intelligent systems**



Message to Students

- **Your Career Journey**
- **You Have Multiple Paths**
- You can become:
 - A researcher (MS/PhD)
 - An AI engineer
 - A systems builder
 - An innovator / entrepreneur
- But success depends on: Depth of understanding, not titles



Message to Students

- **My Advice to You**
- **As Your Instructor**
- If I could give you only a few lessons:
 - Do not chase hype
 - Focus on fundamentals
 - Build things from scratch
 - Stay curious
- And most importantly:
 - **Do not be afraid of difficult problems**



Message to Students

This Is Just the Beginning

“Deep Learning is not the destination.

It is a tool to build the future.”



Message to Students

It Was a Pleasure Teaching You

I'm proud of your effort

Stay in touch

Keep learning

Keep building

The future of AI may be built by one of you

